DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 2

Data scientist

Profession 2 of 10 in Development and IT.

What the profession means

A data scientist builds models that predict or classify, from data preparation and experiments through to a tested model that the business can act on.

What professionals do every day

Frames the problem, prepares data, trains and compares models, evaluates them on held-out data, and explains results to non-technical staff.

Why businesses hire them, and what problems they solve

Predictions guide pricing, stock and marketing. A data scientist says how reliable each prediction is.

- Predictions that look good on training data only.
- Models nobody can explain.
- Experiments that cannot show cause.

Skills you need

- Python and pandas.
- Statistics and experiment design.
- Scikit-learn models and evaluation.
- Communication of uncertainty.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Python and pandas	Analysis in code	Free	Free	pandas.pydata.org
scikit-learn	Machine learning models	Free	Free	scikit-learn.org
Kaggle	Datasets and notebooks	Free	Paid features	www.kaggle.com

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Clean a dataset and compute summaries.
Intermediate	Train one classifier with a test set.
Advanced	Compare two models and explain the choice.

Projects by level

- **Beginner:** Churn prediction for a subscription.
- **Intermediate:** Demand forecast for a shop.

- **Advanced:** Lead scoring for a training firm (see the AI and Automation textbook).

Client-ready portfolio project

A lead-scoring study for a fictional training firm: data preparation, a model with a held-out test score, and a plain-language limitation note.

Worked example

Worked example. A model scores 0.86 on held-out data. The report says what that means for sales, and what it cannot show.

Assignment with full solution

ASSIGNMENT

Model with a held-out score

Business problem: A training firm wants to predict who will enrol.

Steps

1. Split the data into training and test sets with a fixed seed.
2. Train one model and report accuracy on the test set.
3. Write three sentences on what the score does and does not show.

Expected output: A test score and the limitation note.

SOLUTION (ANSWER KEY)

The model is scored only on data it did not see. The note states the limit: the score is for this data and period, not a guarantee for every future lead. The lead-scoring code in the AI and Automation textbook (Chapter 4) shows the same method.

Common mistakes

- Testing on the training data.
- Leaking future information into features.
- Reporting accuracy alone when one class is rare.
- No monitoring after launch.

Freelancing opportunities

- Predictive models for small businesses.
- Forecasts and scoring projects.
- Briefs for 'machine learning', 'predictive model', 'data science'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- scikit-learn user guide
- Kaggle Learn

Part of **Master Freelancing: Data Analytics and Data Science**. **Back to the chapter list on skillnavora.in** · **Open the full textbook**. Official links in this chapter were not checked live; confirm before relying on them.